Home Reading 06 — Physical Modelling and State Space

Before Class

Preview Lecture 6

Read Lecture 6: Physical Modelling and State Space.

Textbook Reading

- Modelling session: Ogata, System Dynamics, 4th ed., Section 6-2, pp. 254–255, and Section 6-4, pp. 270–272 (circuit laws and analogies).
- State-space session: Section 5-1, pp. 169–173 (state variables, state equations and outputs).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

Electrical and electromechanical modelling, physical analogies, state variables, natural modes and state-space representation.

Tutorial Preparation

Read Tutorial 6 before class. Be ready to:

- Q1: Identify energy stores, state variables and the selected output.
- Q2: Use complex impedance and construct a mechanical analogue.
- Q3: Write the balance equation for a single water tank.
- Q4: Model a first-order water-pressure chamber.
- Q5: Form state matrices and compare eigenvalues with transfer-function poles.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- How do energy-storage elements help determine the number of state variables?
- What information does a state model retain that one input-output transfer function does not?

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026
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